THE SUPERIOR UNIVERSITY LAHORE

Week 2 (Fall 2026)

Faculty of Computer Science and Information Technology | Section: BSCS

Subject: COAL-Lab | Total Marks: 10 | Instructor: Muhammad Hamza

Name: Muhammad Adnan Rasheed

Roll No: SU92-BSCSM-F25-324

Question 1: Swapping using a temporary register

Task: Move 10 into AX and 20 into BX, swap them using CX as a temporary register (MOV only). Final: AX = 20, BX = 10, CX = 0.

Code:

```
org 100h

MOV AX, 10      ; AX = 10
MOV BX, 20      ; BX = 20

MOV CX, AX      ; CX = 10 (save AX)
MOV AX, BX      ; AX = 20
MOV BX, CX      ; BX = 10

MOV CX, 0       ; CX = 0

RET
```

Explanation:

AX is saved in CX, then BX is copied into AX, then the saved value in CX is copied into BX. At the end CX is cleared with MOV CX, 0.

Final register values:

Register	Final Value
AX	20
BX	10
CX	0
DX	0

Question 2: Swapping without a temporary register

Task: Move 15 into AX and 25 into BX, swap them using arithmetic (no third register). Final: AX = 25, BX = 15, all other registers = 0.

Code:

```
org 100h

MOV AX, 15      ; AX = 15
MOV BX, 25      ; BX = 25

ADD AX, BX      ; AX = 15 + 25 = 40
SUB BX, AX      ; BX = 25 - 40 = -15
NEG BX          ; BX = 15
SUB AX, BX      ; AX = 40 - 15 = 25

RET
```

Explanation:

AX = AX + BX gives the sum 40. Subtracting AX from BX gives -15, and NEG BX turns it into 15 (the old AX). Finally AX = AX - BX gives 25 (the old BX). CX and DX are never used, so they stay 0.

Final register values:

Register	Final Value
AX	25
BX	15
CX	0
DX	0

Question 3: Chained addition across multiple registers

Task: Move 5 into AX, 10 into BX and 15 into CX. Add all three into DX (DX = 30). AX, BX and CX must be 0 at the end.

Code:

```
org 100h

MOV AX, 5       ; AX = 5
MOV BX, 10      ; BX = 10
MOV CX, 15      ; CX = 15

MOV DX, AX      ; DX = 5
ADD DX, BX      ; DX = 15
ADD DX, CX      ; DX = 30

SUB AX, AX      ; AX = 0
SUB BX, BX      ; BX = 0
SUB CX, CX      ; CX = 0

RET
```

Explanation:

DX starts with the value of AX, then BX and CX are added to it, giving 30. Subtracting a register from itself clears it to 0.

Final register values:

Register	Final Value
AX	0
BX	0
CX	0
DX	30

Question 4: Chained subtraction across multiple registers

Task: Move 50 into AX, 10 into BX and 5 into CX. Subtract BX from AX, then CX from the result, and store the answer in DX (DX = 35). AX, BX and CX must be 0 at the end.

Code:

```
org 100h

MOV AX, 50      ; AX = 50
MOV BX, 10      ; BX = 10
MOV CX, 5       ; CX = 5

MOV DX, AX      ; DX = 50
SUB DX, BX      ; DX = 40
SUB DX, CX      ; DX = 35

SUB AX, AX      ; AX = 0
SUB BX, BX      ; BX = 0
SUB CX, CX      ; CX = 0

RET
```

Explanation:

DX starts as 50. Subtracting BX gives 40 and subtracting CX gives 35. AX, BX and CX are then cleared.

Final register values:

Register	Final Value
AX	0
BX	0
CX	0
DX	35

Question 5: Combined swap, addition and subtraction

Task: Move 8 into AX and 12 into BX. Swap them using arithmetic only (AX = 12, BX = 8). Add the swapped AX and BX into CX (CX = 20). Subtract BX from AX and store the result in DX (DX = 4).

Code:

```
org 100h

MOV AX, 8       ; AX = 8
MOV BX, 12      ; BX = 12

ADD AX, BX      ; AX = 20
SUB BX, AX      ; BX = -8
NEG BX          ; BX = 8
SUB AX, BX      ; AX = 12

MOV CX, AX      ; CX = 12
ADD CX, BX      ; CX = 20

MOV DX, AX      ; DX = 12
SUB DX, BX      ; DX = 4

RET
```

Explanation:

The swap works the same way as in Question 2, giving AX = 12 and BX = 8. CX holds their sum (20) and DX holds AX - BX (4).

Final register values:

Register	Final Value
AX	12
BX	8
CX	20
DX	4